

CHAP. 2

SOME METHODS FOR REGRESSION AND CLASSIFICATION

References:

→ from textbook: chap. 3 – Linear regression; chap. 6
Linear model selection and regularization; chap. 4 – Classification

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Chapter 2 - content

- 2.1. Linear regression models and the least squares method
- 2.2. Linear model selection
- 2.3. *Ridge regression*
- 2.4. Principal Components Regression
- 2.5. Partial least squares regression
- 2.6. Logistic regression
- 2.7. Linear Discriminant Analysis

Obs.: topics are numbered as they appear in the reference book.

3.1 Simple Linear Regression

- we are *regressing* Y on X

$$Y \approx \beta_0 + \beta_1 X$$

- From the training data we will have

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

3.1.1 Estimating the Coefficients

- One criteria is to minimize the residual sum of squares (RSS)

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2$$

where

$$e_i = y_i - \hat{y}_i$$

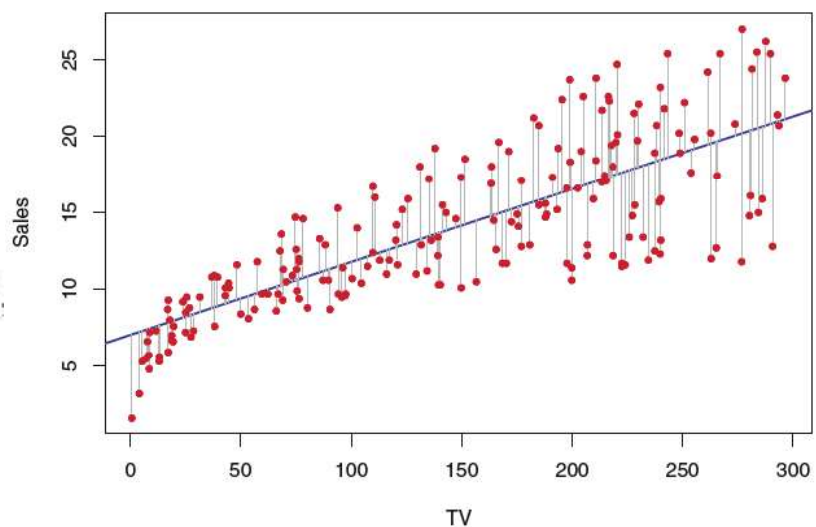


FIGURE 3.1. For the Advertising data, the least squares fit for the regression of sales onto TV is shown. The fit is found by minimizing the sum of squared errors. Each grey line segment represents an error, and the fit makes a compromise by averaging their squares. In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

- Equivalently

$$\text{RSS} = (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \dots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2$$

- And
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

defines the *least squares coefficient estimates* for simple linear regression.

- In our example $\hat{\beta}_0 = 7.03$ and $\hat{\beta}_1 = 0.0475$

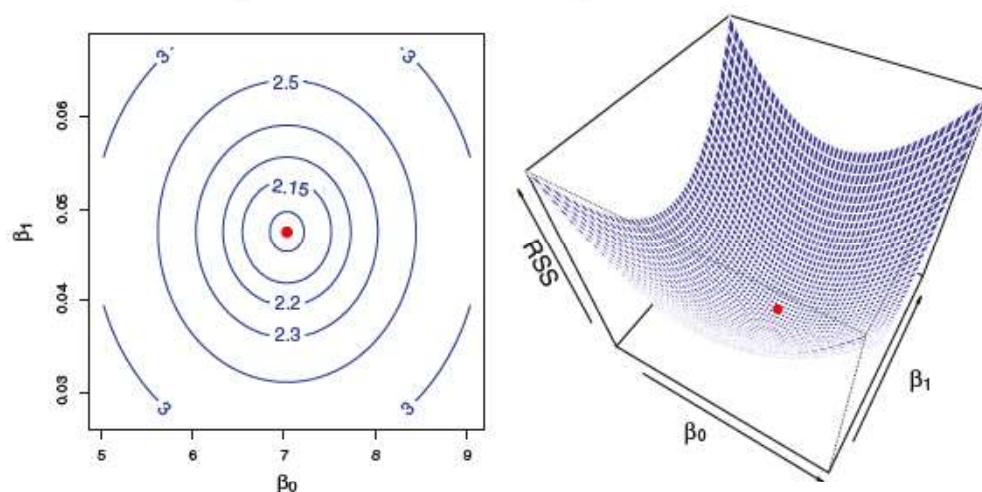


FIGURE 3.2. Contour and three-dimensional plots of the RSS on the Advertising data, using sales as the response and TV as the predictor. The red dots correspond to the least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, given by (3.4).

3.1.2 Assessing the Accuracy of the Coefficient Estimates

- We believe that

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- But each sample could return different values for the coefficients.

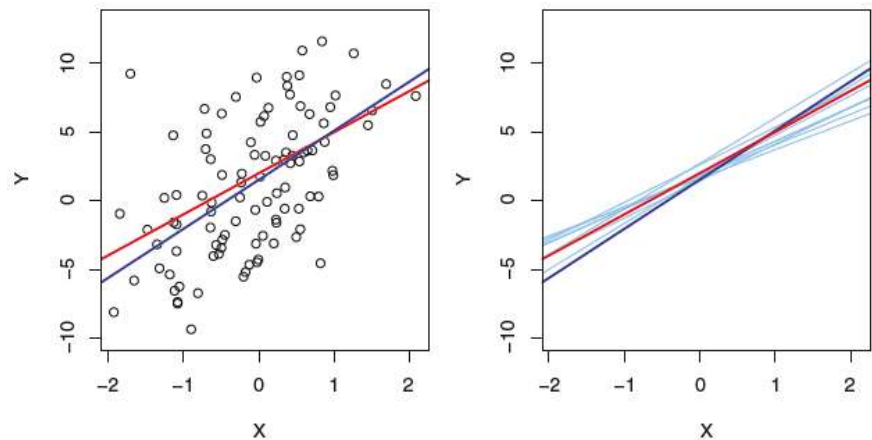


FIGURE 3.3. A simulated data set. Left: The red line represents the true relationship, $f(X) = 2 + 3X$, which is known as the population regression line. The blue line is the least squares line; it is the least squares estimate for $f(X)$ based on the observed data, shown in black. Right: The population regression line is again shown in red, and the least squares line in dark blue. In light blue, ten least squares lines are shown, each computed on the basis of a separate random set of observations. Each least squares line is different, but on average, the least squares lines are quite close to the population regression line.

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased
- And their closeness to the real parameters are given by

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right] \quad SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- Where $\sigma^2 = \text{Var}(\epsilon) \longrightarrow \sigma \longrightarrow \text{RSE} = \sqrt{\text{RSS}/(n-2)}$.
- Assumption: errors are uncorrelated with common variance.

• IC $\hat{\beta}_1 \pm t \cdot SE(\hat{\beta}_1)$ Hip. Test: $\begin{cases} H_0 : \beta_1 = 0 \\ H_a : \beta_1 \neq 0 \end{cases} \quad t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$

$\nwarrow$ $\nearrow$
 $\hat{\beta}_0$

3.1.3 Assessing the Accuracy of the Model

- The quality of a linear regression fit is typically assessed using two related quantities:
- residual standard error (RSE)*

$$\text{RSE} = \sqrt{\frac{1}{n-2} \text{RSS}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- R^2

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$



$$\text{TSS} = \sum (y_i - \bar{y})^2$$

3.2 Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

- 3.2.1 Estimating the Regression Coefficients*
- Least square approach

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2 \end{aligned}$$

- In matrix notation $\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$

- Differentiating w.r.t. β we get the normal equations

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) = 0$$

- If $\mathbf{X}^T\mathbf{X}$ is nonsingular, then the unique solution is given by

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

- and the fitted value at the i th input x_i is

$$\hat{y}_i = \hat{y}(x_i) = x_i^T \hat{\beta}$$

3.2.2 Some Important Questions

- Is There a Relationship Between the Response and Predictors?

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$$

$$H_a : \text{at least one } \beta_j \text{ is non-zero}$$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)}$$

If we want to test that a particular subset of q of the coefficients are zero

$$H_0 : \beta_{p-q+1} = \beta_{p-q+2} = \dots = \beta_p = 0$$

$$F = \frac{(\text{RSS}_0 - \text{RSS})/q}{\text{RSS}/(n - p - 1)}$$

- Deciding on Important Variables
- Variable selection criteria
 - Mallows's Cp; AIC, BIC, Adj_R2
 - Residual plots
 - Automated approach for variable selection (forward, backward, step-wise) obs.: literature shows many others.

- Model Fit:
$$\text{RSE} = \sqrt{\frac{1}{n - p - 1} \text{RSS}} \quad R^2 = \text{Cor}(Y, \hat{Y})^2$$
 - plots

- Predictions: how good is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$
to predict $f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$

3.3 Other Considerations in the Regression Model

- 3.3.1 *Qualitative Predictors*
 - Dummy variable can be used for either two or more qualitative predictors
- See text-book. Pages 82-86

3.3.2 Extensions of the Linear Model

- Linear regression assumes that
 - Relationship between X and Y are **additive** and **linear**
- **Additive**: the effect of changes in a predictor X_j on the response Y is independent of the values of the other predictors.
- **Linear**: the change in the response Y due to a one-unit change in X_j is constant, regardless of the value of X_j
- **How to extend the linear model?**
 - Include interaction terms
 - Use polynomial regression

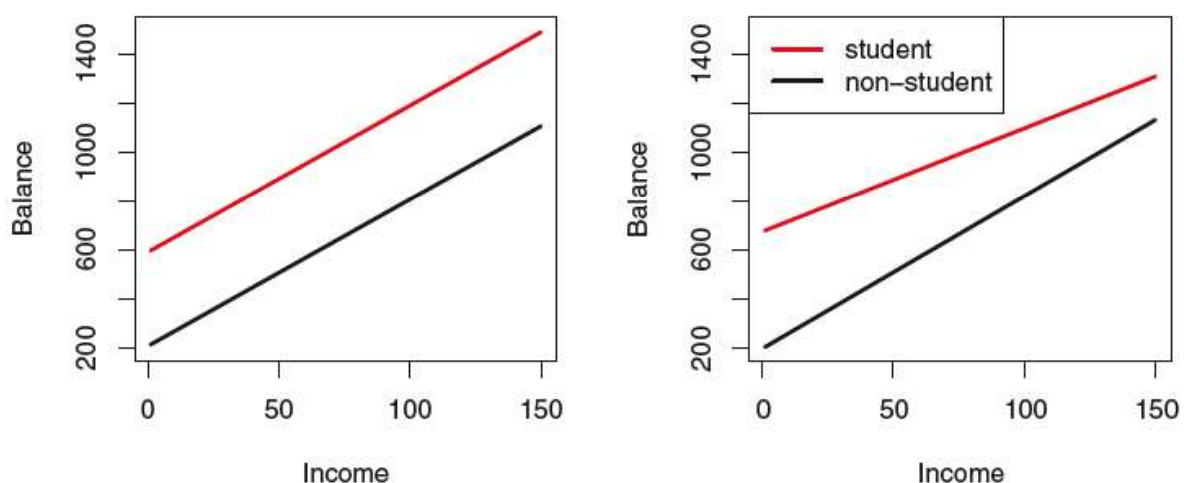


FIGURE 3.7. For the **Credit** data, the least squares lines are shown for prediction of **balance** from **income** for students and non-students. Left: The model (3.34) was fit. There is no interaction between **income** and **student**. Right: The model (3.35) was fit. There is an interaction term between **income** and **student**.

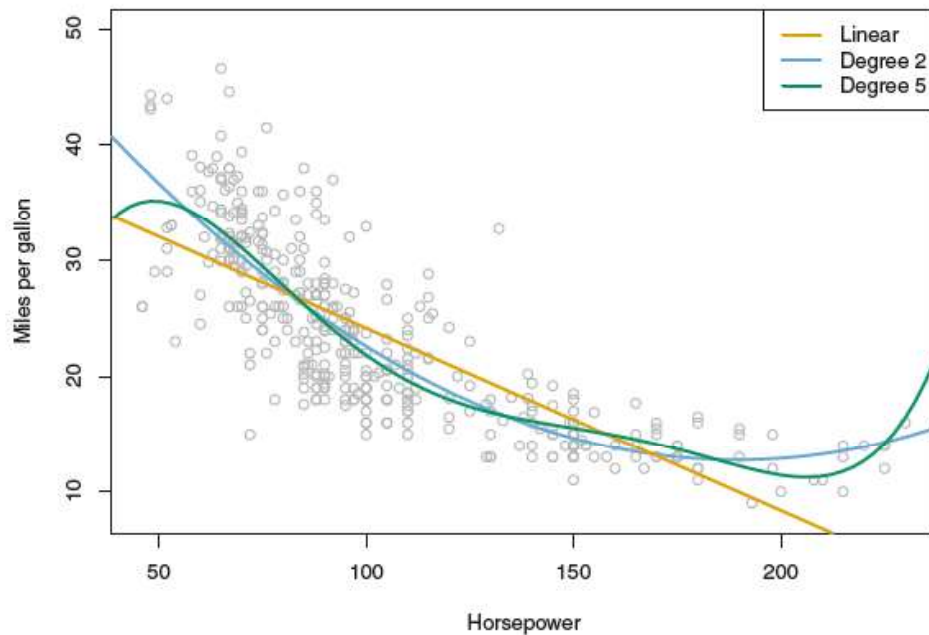


FIGURE 3.8. The *Auto* data set. For a number of cars, *mpg* and *horsepower* are shown. The linear regression fit is shown in orange. The linear regression fit for a model that includes *horsepower*² is shown as a blue curve. The linear regression fit for a model that includes all polynomials of *horsepower* up to fifth-degree is shown in green.

3.3.3 Potential Problems

- Non-linearity of the response-predictor relationships.
- Correlation of error terms.
- Non-constant variance of error terms.
- Outliers.
- High-leverage points.
- Collinearity.
- See textbook for details

1. Non-linearity of the Data

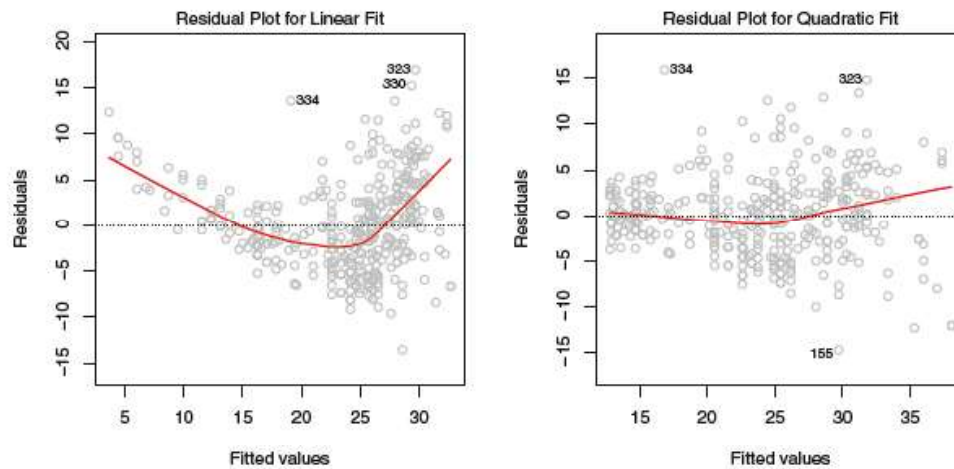


FIGURE 3.9. Plots of residuals versus predicted (or fitted) values for the **Auto** data set. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend. Left: A linear regression of **mpg** on **horsepower**. A strong pattern in the residuals indicates non-linearity in the data. Right: A linear regression of **mpg** on **horsepower** and **horsepower**². There is little pattern in the residuals.

Correlation of Error Terms

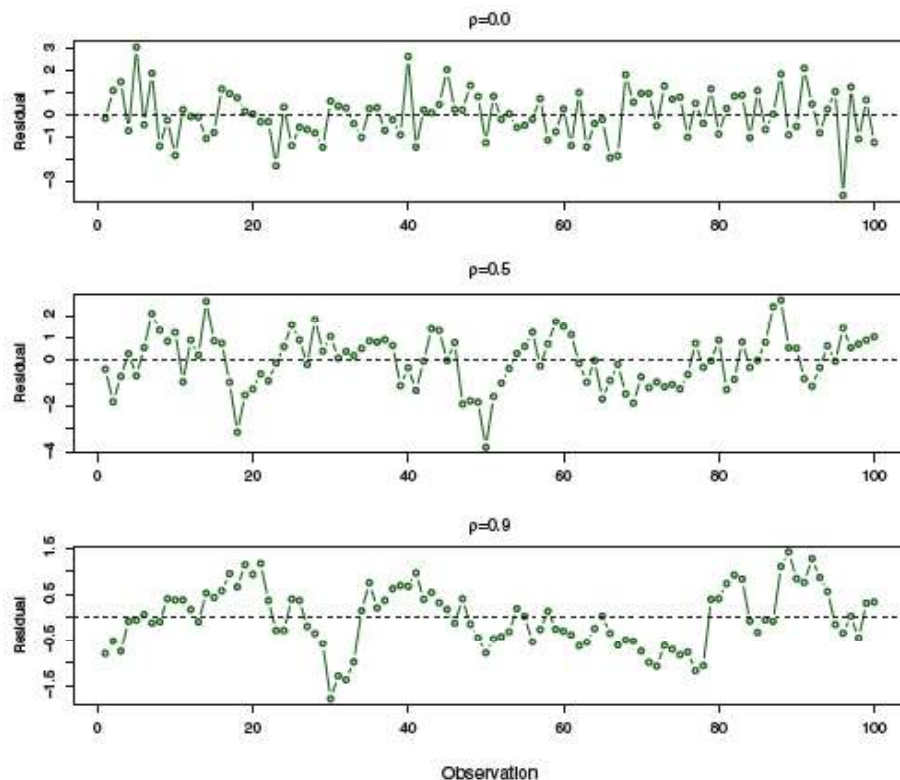


FIGURE 3.10. Plots of residuals from simulated time series data sets generated with differing levels of correlation p between error terms for adjacent time points.

3. Non-constant Variance of Error Terms

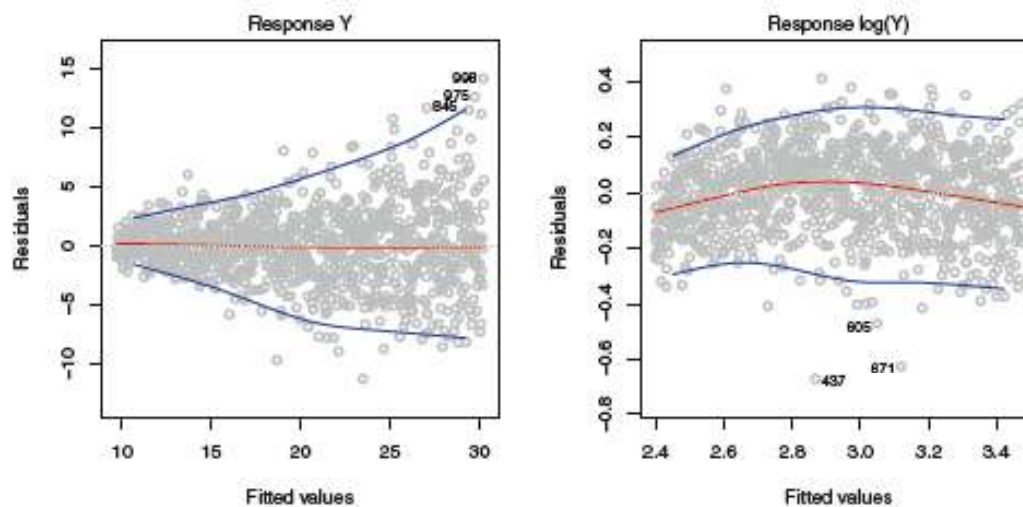


FIGURE 3.11. Residual plots. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend. The blue lines track the outer quantiles of the residuals, and emphasize patterns. Left: The funnel shape indicates heteroscedasticity. Right: The response has been log transformed, and there is now no evidence of heteroscedasticity.

4. Outliers

- An *outlier* is a point for which y_i is far from the value predicted by the model.

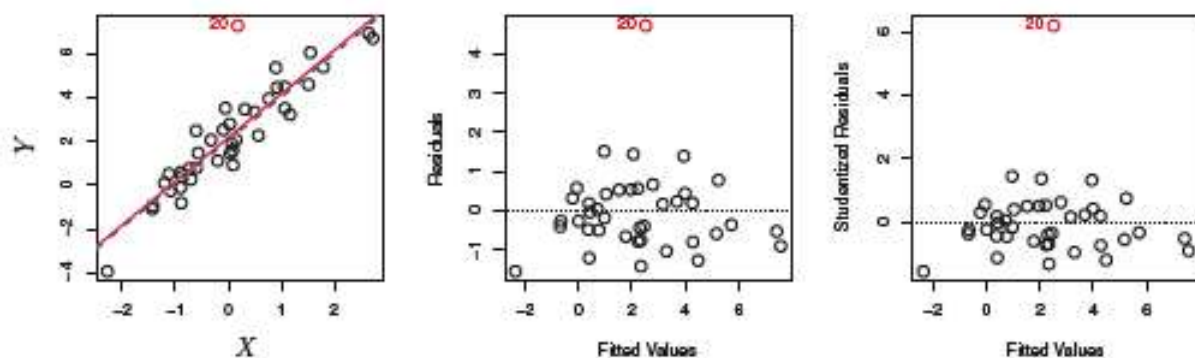


FIGURE 3.12. Left: The least squares regression line is shown in red, and the regression line after removing the outlier is shown in blue. Center: The residual plot clearly identifies the outlier. Right: The outlier has a studentized residual of 6; typically we expect values between -3 and 3 .

5. High Leverage Points

- (outliers: obs for which the response y_i is unusual given the predictor x_i). In contrast, obs with high leverage have an unusual value for x_i .

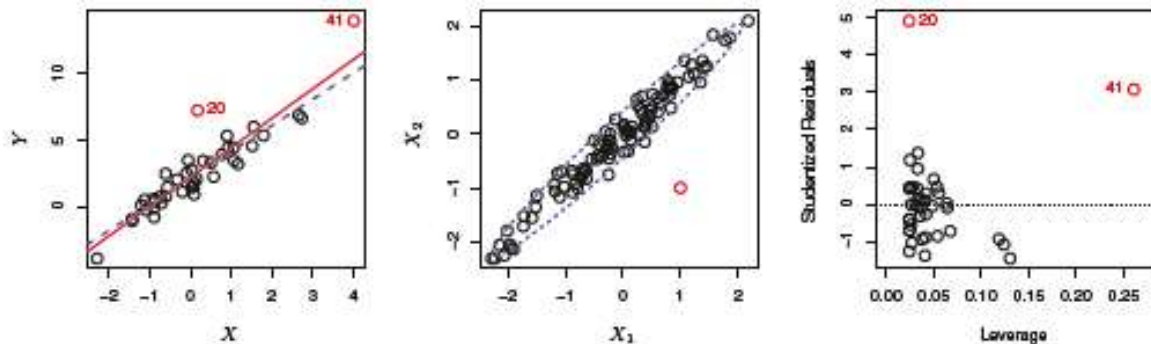


FIGURE 3.13. Left: Observation 41 is a high leverage point, while 20 is not. The red line is the fit to all the data, and the blue line is the fit with observation 41 removed. Center: The red observation is not unusual in terms of its X_1 value or its X_2 value, but still falls outside the bulk of the data, and hence has high leverage. Right: Observation 41 has a high leverage and a high residual.

6. Collinearity

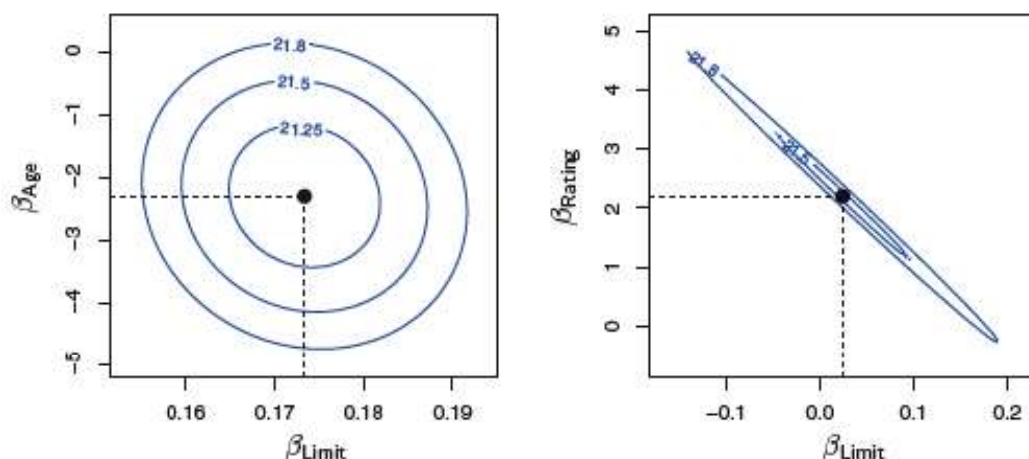


FIGURE 3.15. Contour plots for the RSS values as a function of the parameters β for various regressions involving the **Credit** data set. In each plot, the black dots represent the coefficient values corresponding to the minimum RSS. Left: A contour plot of RSS for the regression of **balance** onto **age** and **limit**. The minimum value is well defined. Right: A contour plot of RSS for the regression of **balance** onto **rating** and **limit**. Because of the collinearity, there are many pairs $(\beta_{\text{Limit}}, \beta_{\text{Rating}})$ with a similar value for RSS.

3.6 Lab: Linear Regression

References:

→ from textbook:

cap. 6 Linear model selection and regularization

6. Introduction

- Linear versus non-linear models
- Linear model has distinct advantages in terms of inference and, on real-world problems, is often surprisingly competitive in relation to non-linear methods.
- we discuss in this chapter some ways in which the simple linear model can be improved, by replacing plain least squares fitting with some alternative fitting procedures.
- Alternative fitting procedures can yield better
 - *prediction accuracy*. (smaller variance $\rightarrow$ better prediction)
 - *model interpretability*. (exclude irrelevant variables)

- There are many alternatives, both classical and modern, to using least squares to fit

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- Three important classes of methods:
 - Subset selection
 - Shrinkage
 - Dimension Reduction

- **Subset Selection.**
 - identifying less than p predictors that we believe to be related to the response → fit a model using LS on the reduced set of variables.
- **Shrinkage.** (also known as *regularization*)
 - fitting a model with p predictors. However, the estimated coefficients are shrunk towards zero relative to the LS estimates. It has the effect of reducing variance. In some cases can also perform variable selection.
- **Dimension Reduction.**
 - *projecting* the p predictors into a M -dimensional subspace ($M < p$). This is achieved by computing M different *linear combinations*, or *projections*, of the variables. These M projections are used as predictors to fit a linear regression model by least squares.

6.1 Subset Selection

- In this section we consider some methods for selecting subsets of predictors.
- These include
 - best subset
 - stepwise model selection procedures.

6.1.1 Best Subset Selection

Algorithm 6.1 Best subset selection

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
 2. For $k = 1, 2, \dots, p$:
 - (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
 - (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .
-
- Up to step 2: low RSS or a high R^2 indicates a model with a low *training* error, whereas we wish to choose a model that has a low *test* error $\rightarrow$ so step 3.

6.1.2 Stepwise Selection

- For computational reasons, best subset selection cannot be applied with very large p .
- It may also suffer from statistical problems for large p .
- The larger the search space, the higher the chance of finding models that look good on the training data, but might not have any predictive power on future data.
- Thus an enormous search space can lead to
 - Overfitting
 - high variance of the coefficient estimates.
- For both of these reasons, *stepwise* methods, which explore a far more restricted set of models, are preferred.

Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
 2. For $k = 0, \dots, p - 1$:
 - (a) Consider all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
 - (b) Choose the *best* among these $p - k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .
-

Backward Stepwise Selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
 2. For $k = p, p - 1, \dots, 1$:
 - (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k - 1$ predictors.
 - (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2 .
 3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2 .
-

Hybrid Approaches

- The best subset, forward stepwise, and backward stepwise selection approaches generally **give similar but not identical models**.
- hybrid versions of forward and backward stepwise selection are alternative
 - variables are added to the model sequentially, in analogy to forward selection. However, after adding each new variable, the method may also remove any variables that no longer provide an improvement in the model fit.
 - Such an approach attempts to more closely mimic best subset selection while retaining the computational advantages of forward and backward stepwise selection.

6.1.3 Choosing the Optimal Model

- Best subset selection, forward selection, and backward selection result in the creation of a set of models, each of which contains a subset of the p predictors
- Respective RSS and R^2 are related to the training error.
- But we wish to choose a model with a low test error.
- training error can be a poor estimate of the test error.
- RSS and R^2 are not suitable for selecting the best model among a collection of models with different numbers of predictors.

- There are two common approaches to estimate the test error :
- 1) indirectly estimate test error by making an *adjustment* to the training error to account for the bias due to overfitting.
- 2) *directly* estimate the test error, using either a validation set approach or a cross-validation approach.
- OBS.: training set MSE is generally an underestimate of the test MSE. Also, $\text{MSE} = \text{RSS}/n$

- **Case 1:** formulas for the case of a linear model fit using least squares

$$C_p = \frac{1}{n} (\text{RSS} + 2d\hat{\sigma}^2)$$

$$\text{AIC} = \frac{1}{n\hat{\sigma}^2} (\text{RSS} + 2d\hat{\sigma}^2)$$

$$\text{BIC} = \frac{1}{n\hat{\sigma}^2} (\text{RSS} + \log(n)d\hat{\sigma}^2)$$

Smaller is better

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n - d - 1)}{\text{TSS}/(n - 1)}$$

Case 2:

Run Validation and Cross-Validation

Select the model for which the estimated test error is smallest

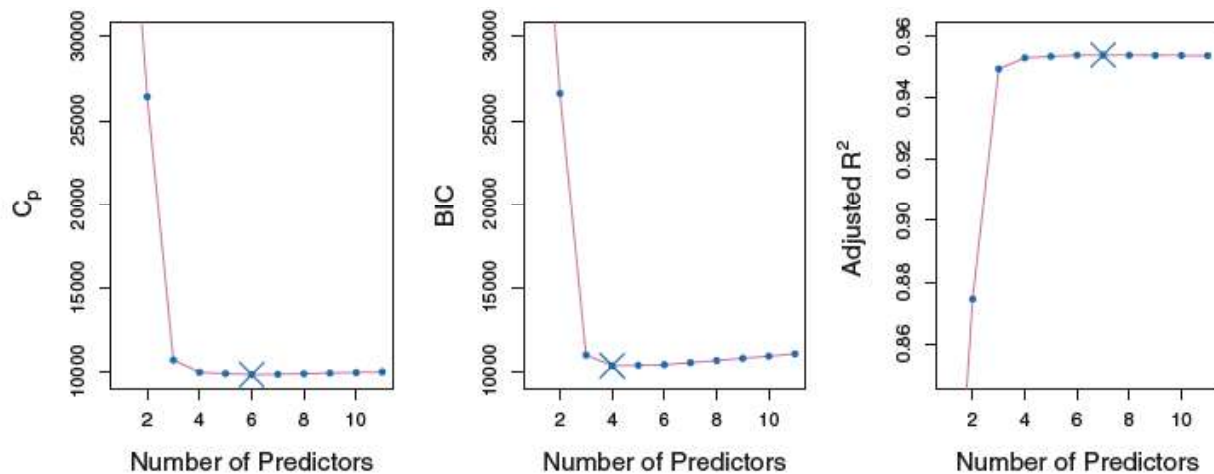


FIGURE 6.2. C_p , BIC, and adjusted R^2 are shown for the best models of each size for the **Credit** data set (the lower frontier in Figure 6.1). C_p and BIC are estimates of test MSE. In the middle plot we see that the BIC estimate of test error shows an increase after four variables are selected. The other two plots are rather flat after four variables are included.

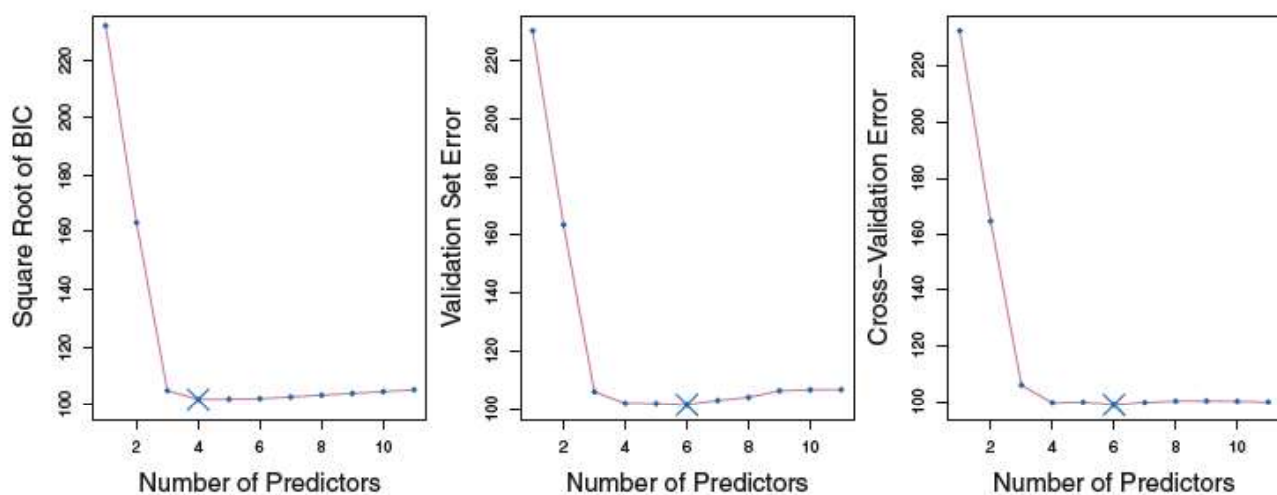


FIGURE 6.3. For the **Credit** data set, three quantities are displayed for the best model containing d predictors, for d ranging from 1 to 11. The overall best model, based on each of these quantities, is shown as a blue cross. Left: Square root of BIC. Center: Validation set errors. Right: Cross-validation errors.

- Using different split of the data, or repeated CV using a different set of CV folds → the precise model with the lowest estimated test error would surely change.
- In this setting, select a model using the *one-standard-error rule*:
 - We first calculate the standard error of the estimated test MSE for each model size
 - then select the smallest model for which the estimated test error is within one standard error of the lowest point on the curve.
- In Figure 6.3 → choose the three-variable model

6.2 Shrinkage Methods

- fit a model containing *all p predictors* using a technique that *constrains* or *regularizes* the coefficient estimates, or equivalently, that *shrinks* the coefficient estimates towards zero.
- shrinking the coefficient estimates can significantly reduce their variance
- The two best-known techniques for shrinking the regression coefficients towards zero are *ridge regression* and the *lasso*.

6.2.1 Ridge Regression

- To find the coefficients that minimize

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

Tuning parameter
Shrinkage penalty

- When $\lambda = 0$ ridge regression will produce the least squares estimates.
- as $\lambda \rightarrow \infty$, ridge regression coefficient estimates will approach zero

Important: to standardize the predictors (same scale)

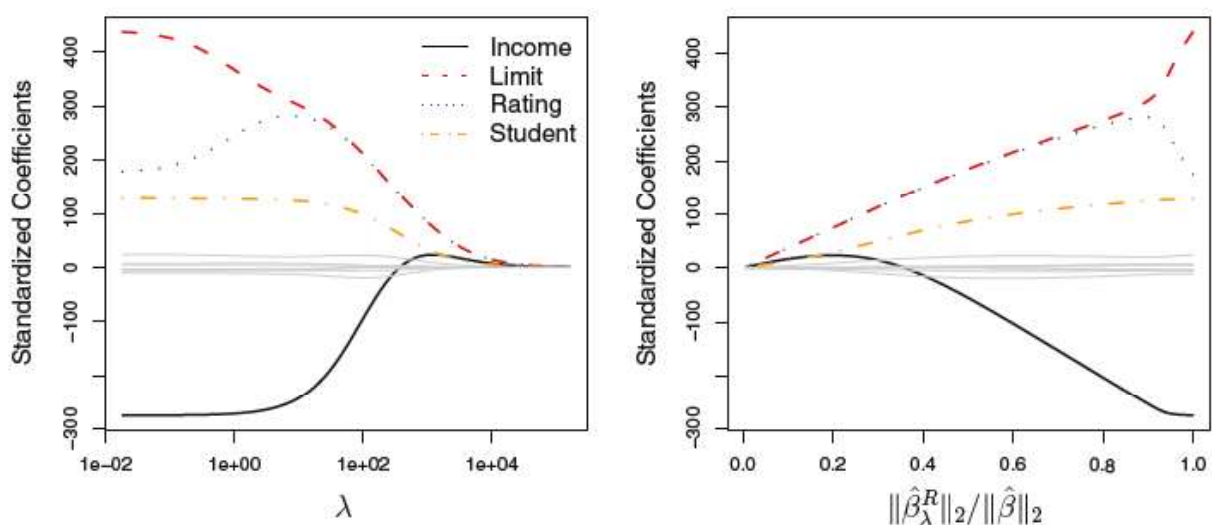


FIGURE 6.4. The standardized ridge regression coefficients are displayed for the **Credit** data set, as a function of λ and $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$.

Measures the distance of β from zero

Why Does Ridge Regression (RR) Improve Over Least Squares?

- Remember the *bias-variance trade-off*.
 - As λ increases, the flexibility of the RR fit decreases, leading to **decreased variance but increased bias**.
- Recall from page 33 (textbook)

$$\underbrace{E \left(y_0 - \hat{f}(x_0) \right)^2}_{\text{Test mean squared error}} = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon)$$

Test mean squared error

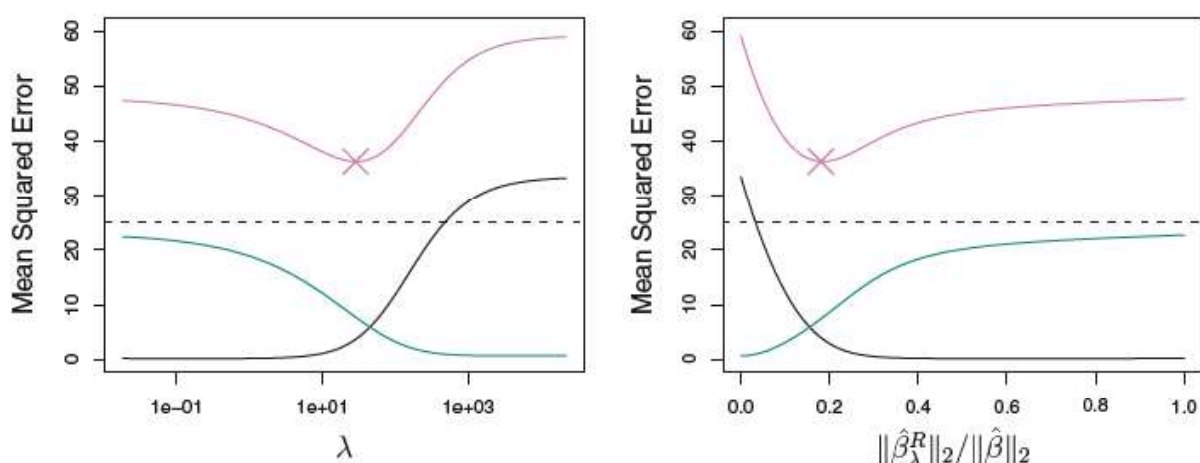


FIGURE 6.5. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set, as a function of λ and $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$. The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.

6.2.3 Selecting the Tuning Parameter

- We choose a **grid of λ** values
- Compute the **cross-validation error for each value of λ** .
- We then **select** the tuning parameter value for which the cross-validation **error is smallest**.
- Finally, the **model is re-fit using all** of the available **observations** and the **selected value of the tuning parameter**

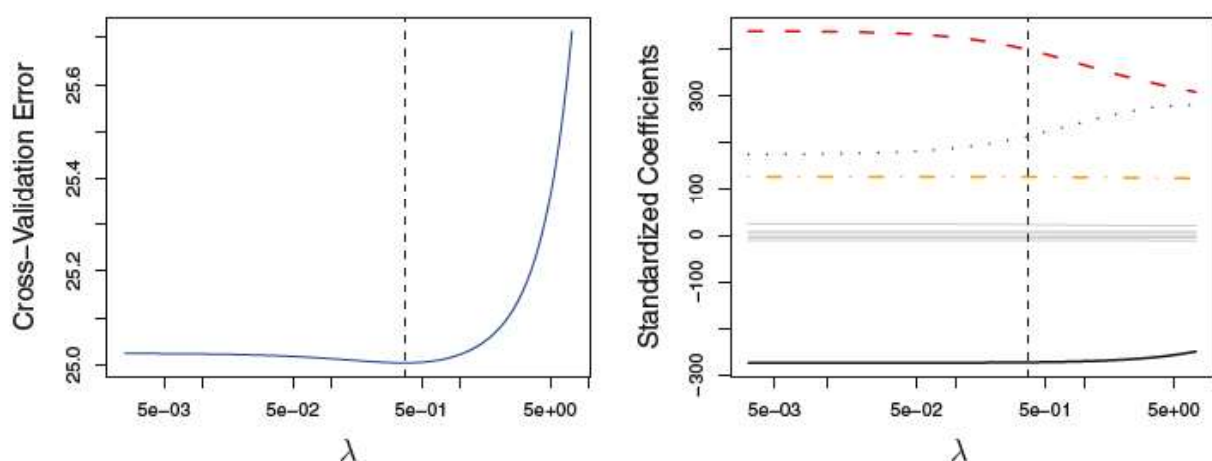


FIGURE 6.12. Left: Cross-validation errors that result from applying ridge regression to the **Credit** data set with various value of λ . Right: The coefficient estimates as a function of λ . The vertical dashed lines indicate the value of λ selected by cross-validation.

6.3 Dimension Reduction Methods

- class of approaches that *transform* the predictors and then fit a least squares model using the transformed variables.
- Let $Z_1, Z_2, \dots, Z_M$ represent $M < p$ linear combinations of our original p predictors

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j$$

- We can then fit the linear regression model

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i$$

- The term *dimension reduction* comes from the fact that this approach reduces the problem of estimating the $p+1$ coefficients $\beta_0, \beta_1, \dots, \beta_p$ to the simpler problem of estimating the $M+1$ coefficients $\theta_0, \theta_1, \dots, \theta_M$, ($M < p$).
- In other words, the dimension of the problem has been reduced from $p+1$ to $M+1$.
- All dimension reduction methods work in two steps.
 - the transformed predictors $Z_1, Z_2, \dots, Z_M$ are obtained.
 - the model is fit using these M predictors.
- the choice of $Z_1, Z_2, \dots, Z_M$, or equivalently, the selection of the ϕ_{jm} 's, can be achieved in different ways.
- In this chapter, we will consider two approaches for this task: *principal components* and *partial least squares*.

6.3.1 Principal Components Regression

- *Principal components analysis* (PCA) is a popular approach for deriving a low-dimensional set of features from a large set of variables.
- PCA is discussed in greater detail as a tool for *unsupervised learning* in Chapter 10 (textbook).
- Here we describe its use as a dimension reduction technique for regression.

An Overview of Principal Components Analysis

- PCA : technique for reducing the dimension of a $n \times p$ data matrix $\mathbf{X}$.
 - The *first PC* (Z_1) direction of the data is that along which the observations *vary the most*.
 - The second PC Z_2 is a linear combination of the variables that is uncorrelated with Z_1 , and has largest variance subject to this constraint
- We will choose M PC such that $M \ll p$.

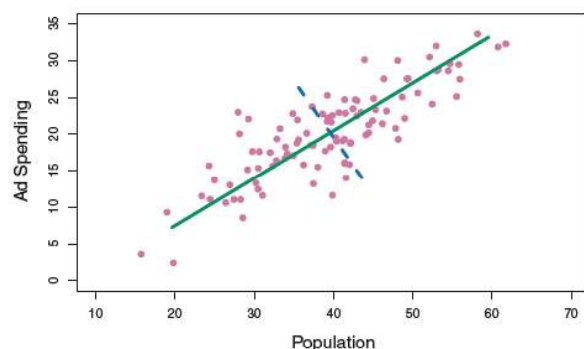


FIGURE 6.14. The population size (*pop*) and ad spending (*ad*) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component, and the blue dashed line indicates the second principal component.

The Principal Components Regression Approach

- PCR approach involves constructing the first M principal components, $Z_1, \dots, Z_M$, and then using these components as the predictors in a **linear regression model**
 - that is fit using least squares
- If having only **two predictors**:

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i$$

$$Z_1 = 0.839 \times (\text{pop} - \overline{\text{pop}}) + 0.544 \times (\text{ad} - \overline{\text{ad}})$$

- If $n = 100$, **pop** and **ad** are vectors of length 100, and so is Z_1

$$z_{i1} = 0.839 \times (\text{pop}_i - \overline{\text{pop}}) + 0.544 \times (\text{ad}_i - \overline{\text{ad}})$$

PC scores

- Also,

$$Z_2 = 0.544 \times (\text{pop} - \overline{\text{pop}}) - 0.839 \times (\text{ad} - \overline{\text{ad}})$$

- **key idea:**
 - often a small number of PC suffice to **explain most of the variability** in the data, as well as the relationship with the response
- **Assumption:** the directions in which $X_1, \dots, X_p$ show the most variation are the directions that are associated with Y .
- **Obs.:** this assumption is not guaranteed to be true, but is a reasonable enough approximation to give good results.

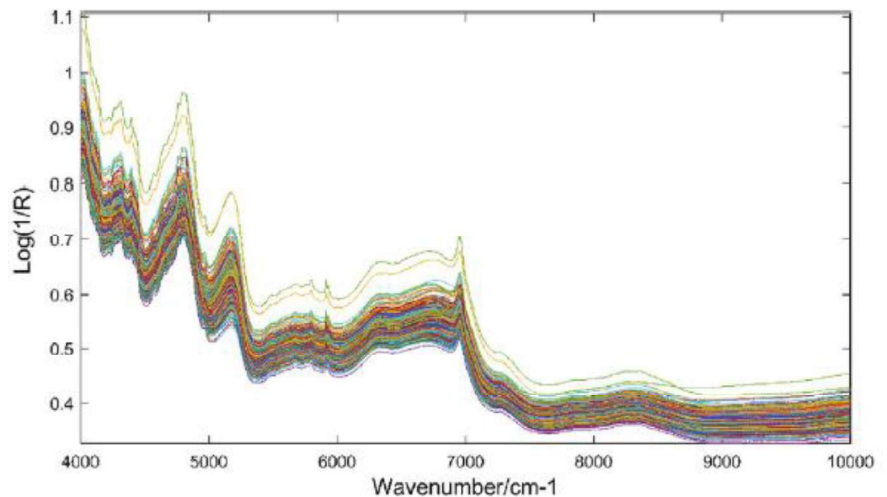
- If the **assumption** underlying PCR holds, then fitting a least squares model to $Z_1, \dots, Z_M$ will lead to **better results** than fitting a least squares model to $X_1, \dots, X_p$, since most or all of the information in the data that relates to the response is contained in $Z_1, \dots, Z_M$, and by estimating only $M \ll p$ coefficients we can **mitigate overfitting**.
- In PCR, the number of PC, M , is typically chosen by CV
- **Standardizing** each predictor is recommended, unless they are on the same units.

6.3.2 Partial Least Squares

- The PCR $\rightarrow$ involves identifying linear combinations, or *directions*, that best represent the predictors $X_1, \dots, X_p$.
 - But, the response does **not supervise** the identification of the principal components.
 - Drawback $\rightarrow$ there is no guarantee that the directions that best explain the predictors will also be the best directions to use for predicting the response.
- PLS is a **supervised** alternative to PCR
- **PLS** identifies these new features in a supervised way
 - **use of the response Y to identify new features** that not only approximate the old features well, but also that *are related to the response*.
 - Roughly speaking, the PLS attempts to find directions that help explain both the response and the predictors.

- The **number M** of partial least squares directions used in PLS is a tuning parameter that is typically chosen by **CV**.
- We generally **standardize** the predictors and response before performing PLS.
- PLS is popular in the field of **chemometrics**, where many variables arise from digitized spectrometry signals.

- In practice it often performs no better than ridge regression or PCR.



Sugarcane samples raw spectra
<https://doi.org/10.1371/journal.pone.0236853.g001>

6.5 Lab 1: Subset Selection Methods

6.6 Lab 2: Ridge Regression

6.7 Lab 3: PCR and PLS Regression

References:

→ from textbook: cap. 4 – Classification

Introduction

- Y (response variable) is qualitative (categorical)
- Process of predicting $Y \rightarrow$ Classification
- Classifiers (classification techniques)
 - Logistic regression
 - Linear discriminant analysis
 - K-nearest neighbors Most widely-used
- Other computer-intensive methods: (examples)
Generalize additive models; trees, random forests; support vector machines

4.1 An Overview of Classification

- Classification problems are (maybe) more often than regression problems
- Examples:
 - A person arrives at the emergency room with a set of symptoms that could possibly be attributed to one of three medical conditions. Which of the three conditions does the individual have?
 - On the basis of DNA sequence data for a number of patients with and without a given disease, a biologist would like to figure out which DNA mutations are deleterious (disease-causing) and which are not.
- Training observations → build a classifier
- Testing observations → check its performance

- In this chapter: uses the simulated **Default** data set
- **Interest:** predicting default on credit card payment as a function of annual income and monthly balance.

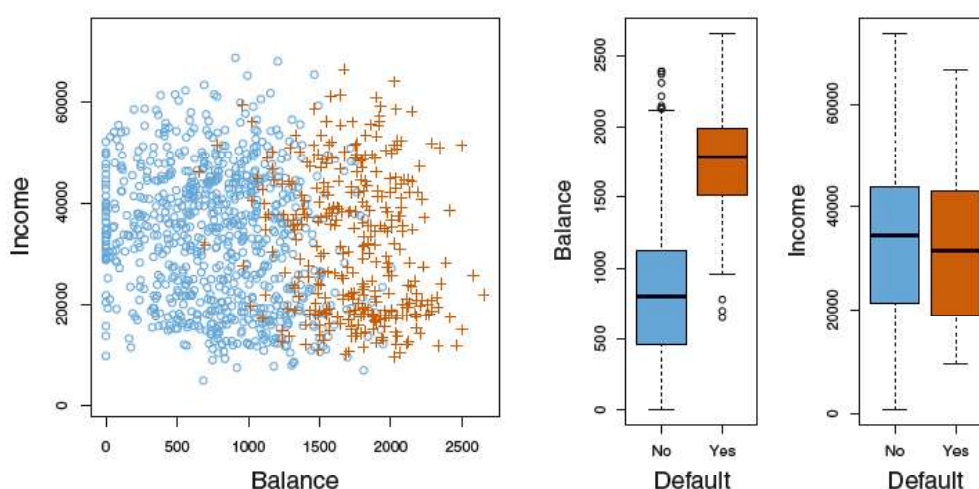


FIGURE 4.1. The **Default** data set. Left: The annual incomes and monthly credit card balances of a number of individuals. The individuals who defaulted on their credit card payments are shown in orange, and those who did not are shown in blue. Center: Boxplots of **balance** as a function of **default** status. Right: Boxplots of **income** as a function of **default** status.

4.2 Why Not Linear Regression?

- Uses a few examples to convince us about avoiding the use of linear regression on classification problems.

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases} \quad Y = \begin{cases} 1 & \text{if epileptic seizure;} \\ 2 & \text{if stroke;} \\ 3 & \text{if drug overdose.} \end{cases}$$

- $\neq$ codings $\rightarrow$ different linear models $\rightarrow$ different sets of predictions on test observations.
- If Y values take on a natural ordering (*mild*, *moderate*, and *severe*), and we felt the gap between levels are similar, then a 1, 2, 3 coding would be reasonable.
- Unfortunately, hard to find a natural way to convert a qualitative response variable with more than two levels into a quantitative response that is ready for linear regression.

- For a *binary* (two level) qualitative response, the situation is better.
- Ex.: use the *dummy variable* approach to code Y:

$$Y = \begin{cases} 0 & \text{if stroke;} \\ 1 & \text{if drug overdose} \end{cases}$$

- fit a linear regression $\rightarrow$ predict drug overdose if $\hat{Y} > 0.5$ and stroke otherwise.

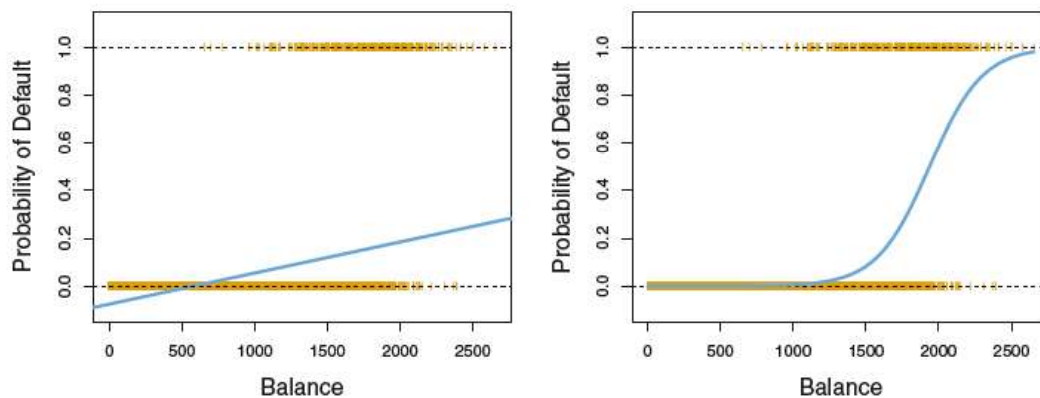


FIGURE 4.2. Classification using the `Default` data. Left: Estimated probability of `default` using linear regression. Some estimated probabilities are negative! The orange ticks indicate the 0/1 values coded for `default` (No or Yes). Right: Predicted probabilities of `default` using logistic regression. All probabilities lie between 0 and 1.

4.3 Logistic Regression

- logistic regression models the *probability* that Y belongs to a particular category

$$\Pr(\text{default} = \text{Yes} | \text{balance}), \quad \text{or} \quad p(\text{balance})$$

- For any given value of balance, a prediction can be made for default. For example:
 - one might predict default = Yes for any individual for whom $p(\text{balance}) > 0.5$.
 - Alternatively, if a company wishes to be conservative in predicting individuals who are at risk for default, then they may choose to use a lower threshold, such as $p(\text{balance}) > 0.1$.

4.3.1 The Logistic Model

- Basic idea:

to model the relationship between $p(X) = \Pr(Y = 1|X)$ and X

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

For any X , $p(X) \in [0, 1]$

Logistic function

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X}$$

Odds ratio $\in [0, \infty]$

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X$$

logit

4.3.2 Estimating the Regression Coefficients

- The coefficients β_0 and β_1 must be estimated based on the available **training data**
- Maximum likelihood is a very general approach that is used to fit many of the non-linear models. It has better statistical properties.

	Coefficient	Std. error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001

TABLE 4.1. For the **Default** data, estimated coefficients of the logistic regression model that predicts the probability of **default** using **balance**. A one-unit increase in **balance** is associated with an increase in the log odds of **default** by 0.0055 units.

- Interpretation: following Table 4.1, p. 134 textbook.

4.3.3 Making Predictions

- Once the coefficients have been estimated → prediction
Ex.: predict that the default probability for an individual with a balance of \$1, 000 is

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1,000}}{1 + e^{-10.6513 + 0.0055 \times 1,000}} = 0.00576$$

- If $X = 2,000 \rightarrow$ default prob = 0.586 (see Figure 4.2)

- Predictor as dummy variable: (if qualitative predictors)

	Coefficient	Std. error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	<0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

TABLE 4.2. For the **Default** data, estimated coefficients of the logistic regression model that predicts the probability of **default** using student status. Student status is encoded as a dummy variable, with a value of 1 for a student and a value of 0 for a non-student, and represented by the variable **student[Yes]** in the table.

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{Yes}) = \frac{e^{-3.5041 + 0.4049 \times 1}}{1 + e^{-3.5041 + 0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{No}) = \frac{e^{-3.5041 + 0.4049 \times 0}}{1 + e^{-3.5041 + 0.4049 \times 0}} = 0.0292.$$

4.3.4 Multiple Logistic Regression

- Predicting a binary response using multiple predictors

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$\searrow$
 $X = (X_1, \dots, X_p)$

- So,

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

- we use the maximum likelihood method to estimate $\beta_0, \beta_1, \dots, \beta_p$.

Example:

	Coefficient	Std. error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	<0.0001
balance	0.0057	0.0002	24.74	<0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

TABLE 4.3. For the **Default** data, estimated coefficients of the logistic regression model that predicts the probability of **default** using **balance**, **income**, and student status. Student status is encoded as a dummy variable **student[Yes]**, with a value of 1 for a student and a value of 0 for a non-student. In fitting this model, **income** was measured in thousands of dollars.

- See textbook for further discussion.
- Problems of “confounding”. Single vs multiple predictors

Making predictions

- a student with a credit card balance of \$1, 500 and an income of \$40, 000 has an estimated probability of default of

$$\hat{p}(X) = \frac{e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 1}}{1 + e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 1}} = 0.058$$

- A non-student with the same balance and income has an estimated probability of default of

$$\hat{p}(X) = \frac{e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 0}}{1 + e^{-10.869+0.00574 \times 1,500+0.003 \times 40-0.6468 \times 0}} = 0.105$$

4.4 Linear Discriminant Analysis (LDA)

- Logistic regression (LR) involves directly modeling

$$\Pr(Y = k|X = x)$$

using the logistic function, given before.

- In LR → we model the conditional distribution of the response Y , given the predictor(s) X .

- LDA is an alternative and less direct approach to estimating $\Pr(Y = k|X = x)$.

- LDA → model the distribution of the predictors X separately in each of the response classes (i.e. given Y), and then use Bayes' theorem to flip these around into estimates for

$$\Pr(Y = k|X = x).$$

Using Bayes' Theorem for Classification

- Suppose that we wish to classify an observation (Y) into one of K classes, where $K \geq 2$.
- Let π_k represent the prior probability that a randomly chosen observation (Y) comes from the k th class;
- Let $f_k(x) \equiv \Pr(X = x|Y = k)$ denote the *density function* of a predictor X for an observ. that comes from the k th class
- Then *Bayes' theorem* states that

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

cont ... $\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)} = p_k(X)$

- So, instead of directly computing $p_k(X)$ as in LR, we can simply plug in estimates of π_k and $f_k(X)$.
- In general, estimating π_k is easy if we have a random sample of Y s from the population:
 - we simply compute the fraction of the training observations that belong to the k th class.
- Estimating $f_k(X)$ tends to be more challenging,
 - unless we assume some simple forms for these densities.
- $p_k(x) \rightarrow$ is the *posterior* probability that an observation $X = x$ belongs to the k th class.
 - That is, it is the probability that the observation belongs to the k th class, *given* the predictor value for that observation.
- the Bayes classifier $\rightarrow$ classifies an observ. to the class for which $p_k(X)$ is largest.
 - It has the lowest possible error rate out of all classifiers. (This is of course only true if the terms above are all correctly specified.)

A simple example

A certain blood bank frequently receives blood samples from donor agencies A, B and C in the ratio of 30:30:40, i.e. 30% from A, 30% from B and 40% from C. From each agency, we know that 10, 5 and 2% of their samples fail the quality control tests. Given a random sample from the bank, answer the following:

- What is the probability that the sample will fail the quality test?
- If the sample failed the quality test, what is the probability it was from agency B?
- If the sample failed the test, which of the three agencies would you complain to?

Solution (in class)

Check file: slides/cap2-bayes_classifier_simple_example

LDA – introd. --

$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)} = p_k(X)$$

- Therefore, if we can find a way to estimate π_k and $f_k(X)$, then we can develop a classifier that approximates the Bayes classifier.
- LDA does this job (if assumptions are met).

4.4.2 Linear Discriminant Analysis for $p = 1$

- Shows an interesting example. See textbook.
- Assume $f_k(X)$ is Gaussian

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right)$$

- Also assume that $\sigma_1^2 = \dots = \sigma_K^2$

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{l=1}^K \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_l)^2\right)}$$

- Taking the log of it we have: $\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$
Bayes classifier

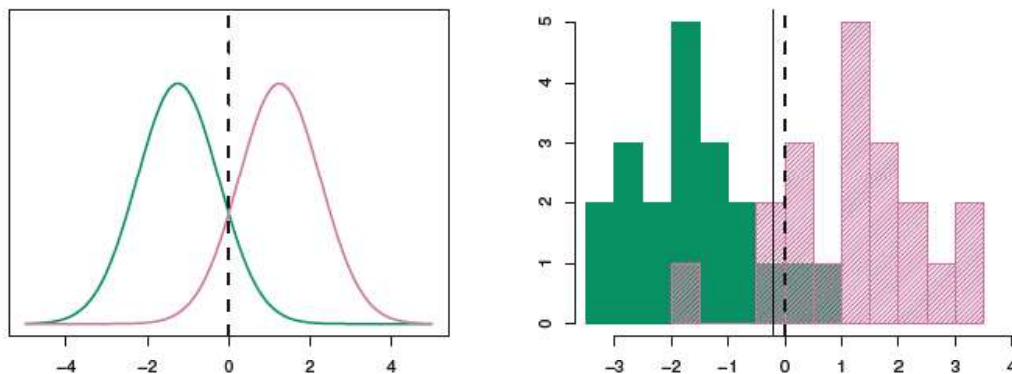


FIGURE 4.4. Left: Two one-dimensional normal density functions are shown. The dashed vertical line represents the Bayes decision boundary. Right: 20 observations were drawn from each of the two classes, and are shown as histograms. The Bayes decision boundary is again shown as a dashed vertical line. The solid vertical line represents the LDA decision boundary estimated from the training data.

Parameter estimates

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i \quad \hat{\pi}_k = n_k/n$$

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

- The **LDA classifier** plugs the estimates into the previous formula and assigns an observation $X = x$ to the class for which

$$\hat{\delta}_k(x) = x \cdot \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k)$$

is largest

4.4.3 Linear Discriminant Analysis for $p > 1$

- assume that $X = (X_1, X_2, \dots, X_p)$ is drawn from a *multivariate Gaussian* distribution, with a class-specific multivariate mean vector and a common covariance matrix.
- Bayes classifier:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

Example (how to simulate it?)

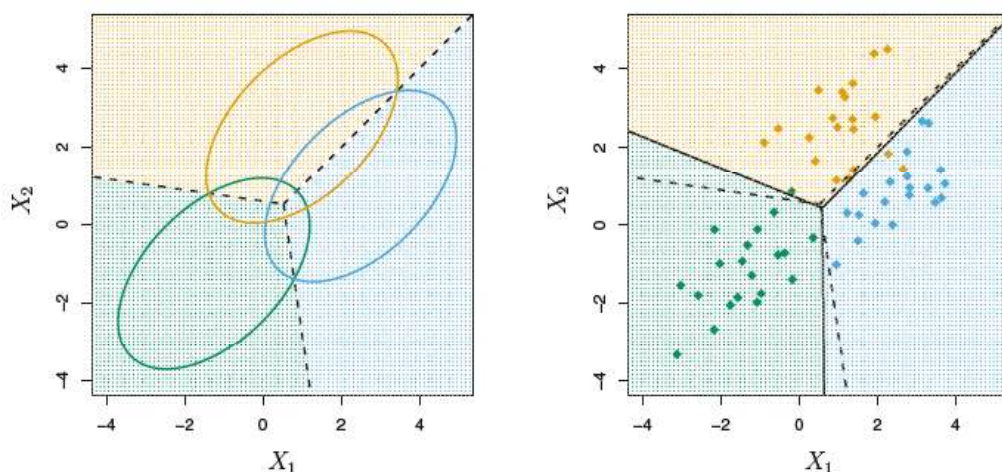


FIGURE 4.6. An example with three classes. The observations from each class are drawn from a multivariate Gaussian distribution with $p = 2$, with a class-specific mean vector and a common covariance matrix. Left: Ellipses that contain 95 % of the probability for each of the three classes are shown. The dashed lines are the Bayes decision boundaries. Right: 20 observations were generated from each class, and the corresponding LDA decision boundaries are indicated using solid black lines. The Bayes decision boundaries are once again shown as dashed lines.

Using Default data set

		True default status		
		No	Yes	Total
Predicted default status	No	9,644	252	9,896
	Yes	23	81	104
Total		9,667	333	10,000

TABLE 4.4. A confusion matrix compares the LDA predictions to the true default statuses for the 10,000 training observations in the **Default** data set. Elements on the diagonal of the matrix represent individuals whose default statuses were correctly predicted, while off-diagonal elements represent individuals that were misclassified. LDA made incorrect predictions for 23 individuals who did not default and for 252 individuals who did default.

- Be careful on overall interpretations!! See p. 145

- The matrix table reveals that LDA predicted that a total of 104 people would default. Of these people, 81 actually defaulted and 23 did not. Hence only 23 out of 9,667 of the individuals who did not default were incorrectly labeled. This looks like a pretty low error rate!
- However, of the 333 individuals who defaulted, 252 (or 75.7%) were missed by LDA. So while the overall error rate is low, the error rate among individuals who defaulted is very high.
- From the perspective of a credit card company that is trying to identify high-risk individuals, an error rate of $252/333 = 75.7\%$ among individuals who default may well be unacceptable.

Medicine and Biology

- the terms *sensitivity* and *specificity* characterize the performance of a classifier or screening test.
- *Sensitivity* is the percentage of true defaulters that are identified, a low 24.3% in this case.
- *Specificity* is the percentage of non-defaulters that are correctly identified, here $(1 - 23 / 9667) \times 100 = 99.8\%$.

- It is possible to *modify the threshold* for classification. Bayes classifier (for $K = 2$) considers it equal to 0.5
- In our example if $\Pr(\text{default} = \text{Yes} | X = x) > 0.2$
- We have:

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9,432	138	9,570
	Yes	235	195	430
	Total	9,667	333	10,000

TABLE 4.5. A confusion matrix compares the LDA predictions to the true default statuses for the 10,000 training observations in the *Default* data set, using a modified threshold value that predicts default for any individuals whose posterior default probability exceeds 20 %.

Trade-off results from modifying the threshold value for the posterior probability of default

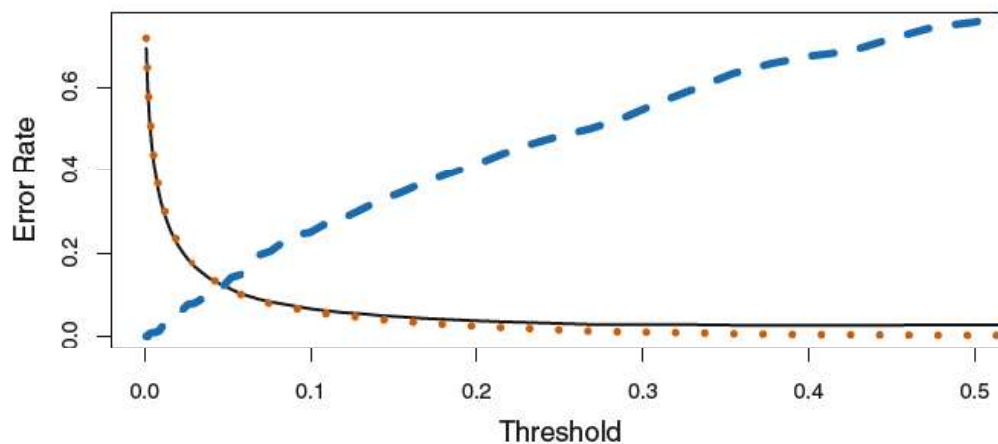


FIGURE 4.7. For the **Default** data set, error rates are shown as a function of the threshold value for the posterior probability that is used to perform the assignment. The black solid line displays the overall error rate. The blue dashed line represents the fraction of defaulting customers that are incorrectly classified, and the orange dotted line indicates the fraction of errors among the non-defaulting customers.

ROC curve

- The *ROC curve* displays, simultaneously, the two types of errors for all possible thresholds.
- The overall performance of a classifier is given by the *area under the (ROC) curve* (AUC).
- The larger the AUC the better the classifier.
- For this data $AUC = 0.95 \rightarrow$ considered very good.
- We expect a classifier that performs no better than chance to have an AUC of 0.5 (when evaluated on an independent **test** set not used in model training).
- ROC curves are **useful for comparing different classifiers**, since they take into account all possible thresholds.

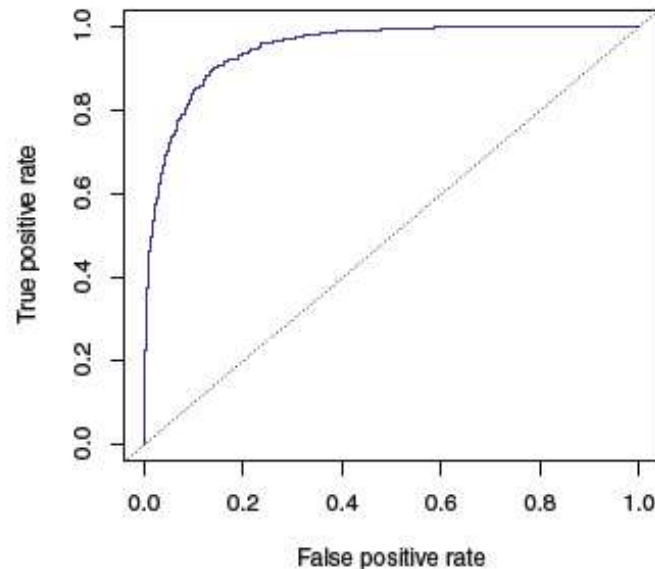


FIGURE 4.8. A ROC curve for the LDA classifier on the **Default** data. It traces out two types of error as we vary the threshold value for the posterior probability of default. The actual thresholds are not shown. The true positive rate is the sensitivity: the fraction of defaulters that are correctly identified, using a given threshold value. The false positive rate is 1-specificity: the fraction of non-defaulters that we classify incorrectly as defaulters, using that same threshold value. The ideal ROC curve hugs the top left corner indicating a high true positive rate and a low false positive rate. The dotted line represents the “no information” classifier; this is what we would expect if student status and credit card balance are not associated with probability of default.

In general

		Predicted class		Total
		– or Null	+ or Non-null	
True class	– or Null	True Neg. (TN)	False Pos. (FP)	N
	+ or Non-null	False Neg. (FN)	True Pos. (TP)	P
Total		N*	P*	

TABLE 4.6. Possible results when applying a classifier or diagnostic test to a population.

Name	Definition	Synonyms
False Pos. rate	FP/N	Type I error, 1–Specificity
True Pos. rate	TP/P	1–Type II error, power, sensitivity, recall
Pos. Pred. value	TP/P*	Precision, 1–false discovery proportion
Neg. Pred. value	TN/N*	

TABLE 4.7. Important measures for classification and diagnostic testing, derived from quantities in Table 4.6.

4.5 A Comparison of Classification Methods

- LDA: $\log \left(\frac{p_1(x)}{1 - p_1(x)} \right) = \log \left(\frac{p_1(x)}{p_2(x)} \right) = c_0 + c_1 x$.
- LR: $\log \left(\frac{p_1}{1 - p_1} \right) = \beta_0 + \beta_1 x$ c_0 and c_1 are functions of μ_1, μ_2 , and σ^2
- So both are linear functions of x . Hence LR and LDA produce linear decision boundaries.
- The only difference between LR and LDA lies in the fact that β_0 and β_1 are estimated using maximum likelihood, whereas c_0 and c_1 are computed using the estimated mean and variance from a normal distribution.
- This same connection between LDA and LR also holds for multidimensional data with $p > 1$.

But not always LR and LDA give similar results

- LDA assumes that the observations are drawn from a Gaussian distribution with a common covariance matrix in each class, and so can provide some improvements over logistic regression when this assumption approximately holds.
- Conversely, logistic regression can outperform LDA if these Gaussian assumptions are not met.

What about KNN? (to be discussed in chapter 4)

- KNN is a completely non-parametric approach:
 - no assumptions are made about the shape of the decision boundary.
- KNN will dominate LDA and logistic regression when the decision boundary is highly non-linear.
- On the other hand, KNN does not tell us which predictors are important; we don't get a table of coefficients as in Table 4.3.

Comparison by simulation – linear scenarios

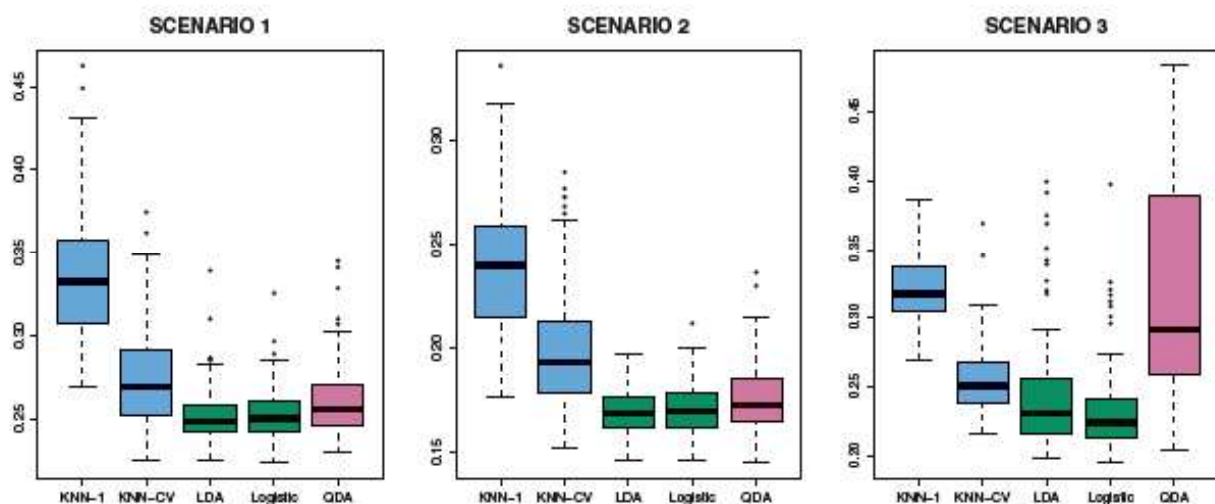


FIGURE 4.10. Boxplots of the test error rates for each of the linear scenarios described in the main text

Comparison by simulation – non-linear scenarios

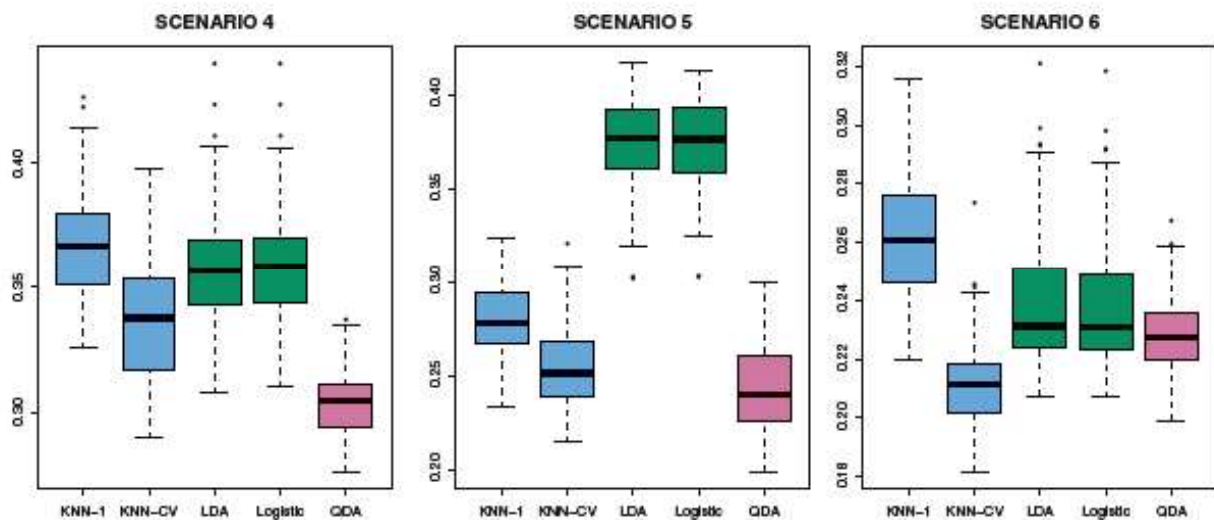


FIGURE 4.11. *Boxplots of the test error rates for each of the non-linear scenarios described in the main text.*

4.6 Lab: Logistic Regression, LDA, QDA, and KNN